

MADHUSOODHANAN SELVAKUMAR

Mechanical Engineer

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SUMMARY

Mechanical Engineering graduate and former ISRO Graduate Apprentice with one year of experience in aerospace structural analysis and technical documentation. Proficient in Siemens NX and Abaqus CAE, with ongoing specialized training in AVEVA E3D. Seeking to apply strong analytical and technical skills in roles across Design, Production, Operations, or Site Engineering.

WORK EXPERIENCE

Graduate Apprentice

Vikram Sarabhai Space Centre (VSSC), ISRO

01/2025 – 01/2026 | Trivandrum, Kerala

- Successfully completed a one-year intensive apprenticeship in the Structural Design and Engineering Division (SDED).
- Assisted in finite element (FE) model assembly and updates for the Gaganyaan Crew Module inner structure.
- Performed meshing, boundary condition setup, and post-processing using Abaqus CAE.
- Collaborated with scientists and engineers to support structural analysis activities.
- Utilized MATLAB for analytical checks and Autodesk Inventor for CAD references.
- Worked on load cases and result interpretation for structural validation
- Gained hands-on expertise in aerospace structural analysis, simulation workflows, and data interpretation.

TECHNICAL SKILLS

- **CAE & Finite Element Analysis:** Abaqus CAE, Ansys Workbench
- **Computer-Aided Design (CAD):** Siemens NX, AutoCAD, Autodesk Inventor, SolidWorks
- **Programming & Mathematical Analysis:** MATLAB, C++, C
- **Product Data Management & Piping:** AVEVA E3D (Training in Progress)
- **Productivity & Design Tools:** MS Office Suite (Word, Excel, PowerPoint), Canva, Adobe Express

EDUCATION

- **Bachelor of Technology (Mechanical Engineering) - CGPA: 8.32**
APJ Abdul Kalam Technological University
2020 – 2024 | Trivandrum, Kerala
- **Senior School Certification (CBSE) – 89.8%**
New Indian Model School
2019 – 2020 | Dubai, UAE

TRAINING/COURSES

- **Training Program in Product Design with Siemens NX**
CADCENTER | 03/2023 – 05/2023
- **Course on Ansys Workbench**
CADCENTER | 09/2023 – 10/2023
- **Training Program on AVEVA E3D**
ASTS Global Education Inc | Present
- **3D Modelling Workshop using Autodesk Fusion 360**
IEDC, College of Engineering Trivandrum | 12/06/2021 – 13/06/2021

ACADEMIC PROJECT

Design and production of door and window jambs from the recyclable plastic waste:

- **Product Development & Prototyping:** Managed the end-to-end design and fabrication workflow for modular construction components using recycled HDPE and PET plastic waste, utilizing Siemens NX for 3D modeling and mold design.
- **Technical Leadership & Analysis:** Served as Team Leader for the research initiative, coordinating material processing and conducting mechanical property evaluations to analyze the structural behavior of recycled composites.

RESPONSIBILITIES HANDLED

- **Volunteer** – NSS CEM | 2022 – 2023
- **Team Leader** – Academic Project | 2023 – 2024

LANGUAGES

English | Malayalam | Tamil | Hindi